

## AI Use Declaration Form

**Course:** Introduction to Artificial Intelligence

**Lab Title:** Lab 1 Part A — Uninformed Search Algorithms

**Student Name:** Meh Ayité Hariel Elinam

**Student ID:** 54392028

**GitHub Repository Link:** [Meh02ajv/ai-course-lab0](https://github.com/Meh02ajv/ai-course-lab0)

**Date Submitted:** Friday, June 5, 2026

### 1. AI Use Summary

| Question                                            | Student Response |
|-----------------------------------------------------|------------------|
| Did you use any AI tool for this lab?               | Yes              |
| Estimated percentage of the work influenced by AI   | 20%              |
| Did you attach evidence of AI use where applicable? | Yes              |

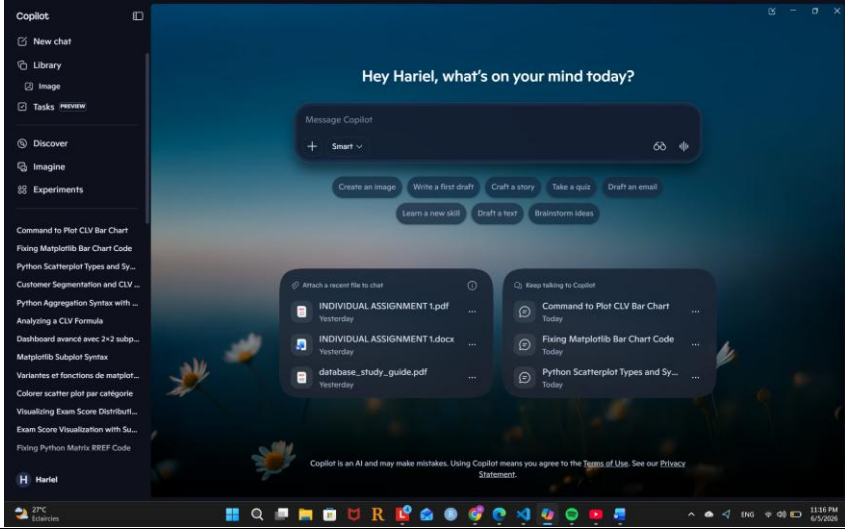
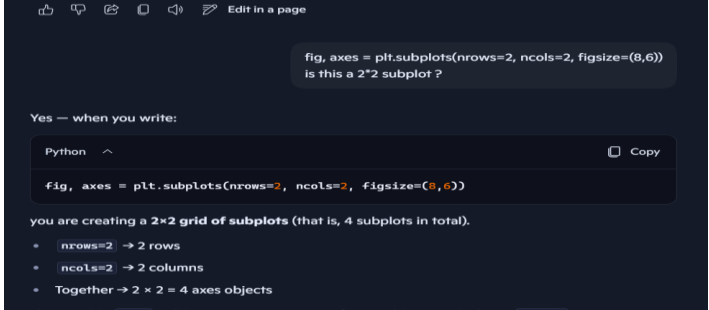
### 2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

| Name of AI Tool Used | Purpose for Using the Tool                                                 | Prompt or Instruction Given to the Tool                                                                                                                                                                                                                                  | Part(s) of the Work Influenced by the Tool | How I Verified, Edited, or Improved the AI Output             |
|----------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------|
| Copilot Assistant    | Correction of Errors, Explanation of concepts, Showing the use of syntaxes | <ul style="list-style-type: none"><li>• type of scatterplot syntax in Python</li><li>• aggregation syntax Python</li><li>• heatmap syntax</li><li>• fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(8,6)) is this a 2*2 subplot ?</li><li>• subplot syntax</li></ul> | Lab0_exercises.ipynb                       | By referring to the lab instructions and Python for ML course |
|                      |                                                                            |                                                                                                                                                                                                                                                                          |                                            |                                                               |
|                      |                                                                            |                                                                                                                                                                                                                                                                          |                                            |                                                               |

### 3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

| Evidence Type                | Attached?<br>(Yes, No, N/A) | File Name                                                                            |
|------------------------------|-----------------------------|--------------------------------------------------------------------------------------|
| AI-generated draft           | No                          |                                                                                      |
| Prompt history               | Yes                         |    |
| Screenshot of AI interaction | Yes                         |  |

|                           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |     | <div> <div>sns.countplot(data=df, x=n_students, hue='year') look</div> <div> The command you wrote has a mistake in the <code>x</code> argument: <div> <div>Python ^</div> <div>Copy</div> </div> <div>sns.countplot(data=df, x=n_students, hue='year')</div> </div> <div> Why it fails <ul style="list-style-type: none"> <li><code>x=n_students</code> → Here you're passing the integer <b>200</b> (since <code>n_students = 200</code> in your dataset creation).</li> <li>But <code>sns.countplot</code> expects a <b>column name (string)</b> from the DataFrame, not a number.</li> </ul> </div> </div> <div> <div>subplot syntax</div> <div> In Matplotlib, the <b>subplot syntax</b> lets you arrange multiple plots in a single figure. Here are the main ways to use it: </div> <div> 1. <code>plt.subplot()</code> <div> <div>Python ^</div> <div>Copy</div> </div> <div> import matplotlib.pyplot as plt<br/> plt.subplot(nrows, ncols, index) </div> <ul style="list-style-type: none"> <li><b>nrows</b>: number of rows in the grid</li> <li><b>ncols</b>: number of columns in the grid</li> <li><b>index</b>: position of the subplot (starts at 1, left to right, top to bottom)</li> </ul> <div>Example:</div> <div> <div>Python ^</div> <div>Copy</div> </div> <div> plt.subplot(1, 2, 1) # 1 row, 2 columns, first plot </div> </div> </div> |
| Exported AI conversation  | No  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| AI-generated code snippet | Yes | <pre> plt.subplot(1, 2, 1) # 1 row, 2 columns, first plot plt.plot([1,2,3], [4,5,6])  plt.subplot(1, 2, 2) # second plot plt.plot([1,2,3], [6,5,4]) plt.show() import matplotlib.pyplot as plt  x = [1, 2, 3, 4, 5] y = [2, 4, 1, 3, 5]  plt.scatter(x, y, c='blue', s=50, marker='o', alpha=0.7) plt.xlabel("X-axis") plt.ylabel("Y-axis") plt.title("Basic Scatterplot") plt.show()  import seaborn as sns import pandas as pd </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                |    |                                                                                                                                                                                                                                                                 |
|------------------------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                |    | <pre>df = pd.DataFrame({     "x": [1,2,3,4,5],     "y": [2,4,1,3,5],     "category": ["A","B","A","C","B"] })  sns.scatterplot(data=df, x="x", y="y", hue="category",     style="category", size="y") plt.title("Scatterplot with Categories") plt.show()</pre> |
| Revised<br>version<br>showing<br>student input | No |                                                                                                                                                                                                                                                                 |

#### 4. Student Declaration

I declare that:

| Declaration Statement                                                                     | Response<br>(Yes/No) |
|-------------------------------------------------------------------------------------------|----------------------|
| The submitted work is my own work.                                                        | Yes                  |
| Any use of AI tools has been clearly declared in this form.                               | Yes                  |
| The prompts, instructions, and AI-generated outputs have been disclosed where applicable. | Yes                  |
| I reviewed, tested, edited, and improved any AI-generated content before submission.      | Yes                  |
| My AI usage does not exceed 25% of the entire work.                                       | Yes                  |
| I understand that undeclared or excessive AI use may be treated as academic misconduct.   | Yes                  |

**Student Signature:** Hariel

**Date:** Friday, June 5, 2026